Development Log

Team 52 - Clue! (G6046 Software Engineering)

This log records individual working sessions, coding progress and decisions made outside of formal team meetings.

Progress Summary

Date	Hours	Key Activity
24 February 2026 (Tuesday)	1.5 hrs	Initial project reading and team setup
26 February 2026 (Thursday)	1 hr	GitHub setup and project structure
28 February 2026 (Saturday)	2 hrs	Class structure research and design planning
02 March 2026 (Monday)	2 hrs	Skeleton Java classes and IntelliJ setup
03 March 2026 (Tuesday)	1 hr	First team meeting - language, roles, ghosts
04 March 2026 (Wednesday)	1.5 hrs	Sprint 1 planning and documentation
05 March 2026 (Thursday)	1.5 hrs	Second team meeting - roles, tech stack, Discord
07 March 2026 (Saturday)	2 hrs	Risk analysis and Sprint 1 document
09 March 2026 (Monday)	1 hr	Partial team meeting - Insik ill, Omar absent
11 March 2026 (Wednesday)	0.5 hrs	Team follow-up attempts
16 March 2026 (Monday)	4 hrs	Core class implementation - catching up after illness
17 March 2026 (Tuesday)	3 hrs	Game class, Suggestion and Accusation implementation
19 March 2026 (Thursday)	2.5 hrs	Sprint 1 testing and wrap-up
22 March 2026 (Sunday)	2 hrs	Sprint 1 review and cleanup
24 March 2026 (Tuesday)	3 hrs	Sprint 2 turn flow and console prototype work
27 March 2026 (Friday)	3.5 hrs	Movement validation and board/gameplay logic
30 March 2026 (Monday)	3 hrs	Suggestion, disproving and accusation flow
02 April 2026 (Thursday)	2.5 hrs	AI player and console game improvements
05 April 2026 (Sunday)	3 hrs	JUnit expansion and Sprint 2 stabilisation
09 April 2026 (Thursday)	2 hrs	Sprint 2 review and handoff planning
12 April 2026 (Sunday)	1 hr	Other coursework and team follow-up
15 April 2026 (Wednesday)	1 hr	TA / lecturer contact and Sprint 3 planning
19 April 2026 (Sunday)	3 hrs	Sprint 3 start - JavaFX setup and board display
22 April 2026 (Wednesday)	4 hrs	GUI sidebar, controls and movement interaction

24 April 2026 (Friday)	4 hrs	Suggestion, accusation dialogs and privacy screen
25 April 2026 (Saturday)	3 hrs	Final gameplay fixes, AI updates and test suite completion
25 April 2026 (Saturday)	3 hrs	Final gameplay fixes, AI updates and test suite completion
26 April 2026 (Sunday)	5 hrs	Documentation - sprint docs, testing doc, design doc and dev log
27 April 2026 (Monday)	4 hrs	Document refinement and submission-pack preparation
28 April 2026 (Tuesday)	4 hrs	Final submission checkpoint, peer review preparation and final documentation pass

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Detailed Log Entries

Entry 1 - 24 February 2026 (Tuesday)

Date	Hours	Focus
24 February 2026 (Tuesday)	1.5 hrs	Initial project reading and team setup

Work completed:

- Read through the full coursework specification and marking criteria carefully.
- Identified the 9 core game requirements and key deliverables - Gantt, risk analysis, sprint docs, group report, video, peer review.
- Noted that 2 team members had not responded to initial team formation emails.
- Sent follow-up email to Shengyuan with Instagram group chat link - no response received.

Blockers / issues:

- Two team members have been completely unresponsive since the project started.

Planned next steps:

- Set up a GitHub repository and arrange a first team meeting.

Entry 2 - 26 February 2026 (Thursday)

Date	Hours	Focus
26 February 2026 (Thursday)	1 hr	GitHub setup and project structure

Work completed:

- Created private GitHub repository: team52-clue-g6046.

- Created initial folder structure: src/com/cluegame/{model,cards,players,gui}, docs/, tests/.
- Added .gitkeep files to track empty folders and pushed initial commit.
- Added README.md with project overview.

Blockers / issues:

- None.

Planned next steps:

- Arrange first team meeting. Begin researching Clue! game rules and class structure.

Entry 3 - 28 February 2026 (Saturday)

Date	Hours	Focus
28 February 2026 (Saturday)	2 hrs	Class structure research and design planning

Work completed:

- Studied the Clue! game rules in detail from the coursework spec.
- Researched Java OO design patterns relevant to card games and board games.
- Drafted initial class hierarchy: identified Card, Player, Board, Room, Square, Dice, Deck, MurderEnvelope, Suggestion, Accusation.
- Decided on package structure: com.cluegame.model, cards, players, gui.
- Note that loading board layout from external config file is required by Watson Games spec.

Blockers / issues:

- None.

Planned next steps:

- Create skeleton Java classes and meet with the team.

Entry 4 - 02 March 2026 (Monday)

Date	Hours	Focus
02 March 2026 (Monday)	2 hrs	Skeleton Java classes and IntelliJ setup

Work completed:

- Configured IntelliJ
- Marked src/ as Sources Root and tests/ as Test Sources Root.
- Created skeleton Java classes for all packages with JavaDoc comments.
- All 17 skeleton classes compile cleanly with Java 21.
- Committed and pushed skeleton codebase to GitHub.

Blockers / issues:

- None.

Planned next steps:

- First team meeting on 3 March. Begin filling in class logic.

Entry 5 - 03 March 2026 (Tuesday)

Date	Hours	Focus
03 March 2026 (Tuesday)	1 hr	First team meeting - language, roles, ghosts

Work completed:

- Met with Insik in person at 12:30 for approximately 30 minutes.
- Agreed on Java as the programming language.
- Discussed Insik taking the Project Lead role - agreed provisionally.
- Discussed approach to the two non-responding members - agreed to proceed without them and document non-response.

Blockers / issues:

- Murat and Shengyuan are still not responding.

Planned next steps:

- Meet with Omar on 5 March to confirm roles and setup.

Entry 6 - 04 March 2026 (Wednesday)

Date	Hours	Focus
04 March 2026 (Wednesday)	1.5 hrs	Sprint 1 planning and documentation

Work completed:

- Drafted initial Sprint 1 document.
- Outlined functional and non-functional requirements for Sprint 1.
- Began risk analysis: identified initial risks around team engagement, technical complexity and scope.

Blockers / issues:

- Roles not yet fully confirmed pending full team agreement.

Planned next steps:

- Meeting with Omar 5 March. Finalise Sprint 1 doc and risk analysis.

Entry 7 - 05 March 2026 (Thursday)

Date	Hours	Focus

05 March 2026 (Thursday)	1.5 hrs	Second team meeting - roles, tech stack, Discord

Work completed:

- Met with Omar in person.
- Agreed on full tech stack: Java 21, IntelliJ IDEA, JUnit 5, GitHub.
- Confirmed preliminary roles: Insik (Project Lead/cards), Thanh (board/movement), Omar (testing/documentation).
- Agreed to set up Discord for team communication.
- Planned full team meeting for Monday 9 March.

Blockers / issues:

- Murat and Shengyuan still absent.

Planned next steps:

- Full team meeting 9 March. Continue Sprint 1 doc and risk analysis.

Entry 8 - 07 March 2026 (Saturday)

Date	Hours	Focus
07 March 2026 (Saturday)	2 hrs	Risk analysis and Sprint 1 document

Work completed:

- Completed full risk analysis with 21 identified risks across team, technical, scope, process, git and submission categories.
- Added risk review schedule tied to sprint dates.
- Updated Sprint 1 document with correct roles, task tables and sprint goal.
- Added functional and non-functional requirements to Sprint 1 doc.

Blockers / issues:

- None.

Planned next steps:

- Team meeting 9 March - confirm roles and start coding.

Entry 9 - 09 March 2026 (Monday)

Date	Hours	Focus
09 March 2026 (Monday)	1 hr	Partial team meeting - Insik ill, Omar absent

Work completed:

- Attended planned team meeting location: waited 50 minutes for Omar who did not arrive.
- Insik contacted via Instagram: confirmed ill, available online only.

- Sent detailed DM to Insik outlining Project Lead responsibilities, class ownership, tech stack and Sprint 1 deadline.
- Attempted to reorganise meeting for later in the week via group chat: no responses received.

Blockers / issues:

- Omar unresponsive. Meeting could not proceed as planned.

Planned next steps:

- Proceed independently on coding. Follow up with team members.

Entry 10 - 11 March 2026 (Wednesday)

Date	Hours	Focus
11 March 2026 (Wednesday)	0.5 hrs	Team follow-up attempts

Work completed:

- Sent message in Instagram group chat attempting to reschedule team meeting: no response.
- Checked Discord: no activity from any team members.
- Noted Murat present in group chat but not responding to messages.

Blockers / issues:

- Zero team engagement.

Planned next steps:

- Continue chasing team members.

Entry 11 - 16 March 2026 (Monday)

Date	Hours	Focus
16 March 2026 (Monday)	4 hrs	Core class implementation - catching up after illness

Work completed:

- Implemented Board class: 25x24 grid, 9 rooms with secret passages, external config file loader.
- Implemented Room class: name, doors, secret passage, occupants list.
- Implemented Square class: row, col, type enum, occupied flag.
- Implemented Dice class: fair two-die roll using java.util.Random.
- Implemented Card abstract class and SuspectCard, WeaponCard, RoomCard subclasses.
- Implemented Deck class: all 21 cards, shuffle and deal logic.
- Implemented MurderEnvelope class: seal() and verify() methods.
- Implemented Player abstract class and HumanPlayer and AIPlayer subclasses.
- All classes include full JavaDoc. Build completed successfully with Java 21.
- Reorganised docs folder.

Blockers / issues:

- Sprint 1 extended to 22 March due to illness. Still no contributions from other team members.

Planned next steps:

- Implement Game class and Suggestion/Accusation logic. Write JUnit tests.

Entry 12 - 17 March 2026 (Tuesday)

Date	Hours	Focus
17 March 2026 (Tuesday)	3 hrs	Game class, Suggestion and Accusation implementation

Work completed:

- Implemented Game.startGame(): shuffles deck, seals envelope with one card of each type, deals remaining 18 cards to players, places tokens at fixed starting positions.
- Implemented Game.nextTurn(): prints current player, advances turn counter, handles active player cycling.
- Implemented Suggestion as a plain value object holding one card of each type.
- Implemented Accusation extending Suggestion: semantic distinction for envelope checking.
- Added main() method to Game for verification: creates 1 HumanPlayer and 1 AIPlayer, runs startGame() and prints hands.
- Verified all 21 cards accounted for: 9 dealt to Alice, 9 to Bob, 3 in envelope.
- All classes compile and build cleanly with Java 21.

Blockers / issues:

- None.

Planned next steps:

- Write JUnit tests. Commit Sprint 1 code.

Entry 13 - 19 March 2026 (Thursday)

Date	Hours	Focus
19 March 2026 (Thursday)	2.5 hrs	Sprint 1 testing and wrap-up

Work completed:

- Added JUnit 5 as a library dependency in IntelliJ.
- Wrote DiceTest: roll() range and rollSingle() range over 1000 iterations.
- Wrote DeckTest: exactly 21 cards, correct type breakdown of 6/6/9.
- Wrote MurderEnvelopeTest: verify() true on match, false on wrong cards, isSealed() before and after.
- Wrote GameTest: envelope sealed after startGame(), each player gets correct cards, no overlap with envelope.
- All 11 JUnit tests passing.
- Committed all Sprint 1 code with message: Sprint 1 complete: Game, Suggestion, Accusation, JUnit tests all passing.

Blockers / issues:

- None.

Planned next steps:

- Tag Sprint 1 release. Begin Sprint 2.

Entry 14 - 22 March 2026 (Sunday)

Date	Hours	Focus
22 March 2026 (Sunday)	2 hrs	Sprint 1 review and cleanup

Work completed:

- Tagged GitHub release v0.1-sprint1.
- Set up Git branching structure: created dev and feature/sprint2 branches.
- Wrote post-sprint review for Sprint 1: noted illness, team issues and what carried over.
- Updated Sprint 1 document with final task statuses.
- Reviewed risk register: noted R01 (ghost members) and R02 (schedule) as materialised.

Blockers / issues:

- None.

Planned next steps:

- Begin Sprint 2 coding: game loop, movement validation, console prototype.

Entry 15 - 24 March 2026 (Tuesday)

Date	Hours	Focus
24 March 2026 (Tuesday)	3 hrs	Sprint 2 turn flow and console prototype work

Work completed:

- Implemented full board layout: 9 rooms in correct positions matching the official Clue! board image.
- Added blocked wall squares, door squares and 6 player starting positions.
- Added central staircase blocked area rows 10-13.
- Implemented secret passages: Study to Kitchen and Conservatory to Lounge.
- Started Game.run() - basic game loop structure.
- Started movement system: HumanPlayer console input and AIPlayer random movement.

Blockers / issues:

- None.

Planned next steps:

- Complete movement validation. Implement suggestion and accusation flow.

Entry 16 - 27 March 2026 (Friday)

Date	Hours	Focus
27 March 2026 (Friday)	3.5 hrs	Movement validation and board/gameplay logic

Work completed:

- Implemented Board.isValidMove(): checks adjacency, bounds, blocked squares, room entry rules and corridor occupancy.
- Implemented HumanPlayer movement: N/S/E/W step-by-step console input with remaining steps shown.
- Implemented AIPlayer movement: Manhattan distance heuristic toward nearest room door.
- Implemented room entry and exit logic: enterRoom() and leaveRoom() on Player.
- Fixed weapon names to match spec: Dagger, Lead Piping, Spanner replacing earlier incorrect names.
- Console game now runs: players take turns, move around the board and enter rooms.

Blockers / issues:

- None.

Planned next steps:

- Implement suggestion and disproving mechanism.

Entry 17 - 30 March 2026 (Monday)

Date	Hours	Focus
30 March 2026 (Monday)	3 hrs	Suggestion, disproving and accusation flow

Work completed:

- Implemented full suggestion flow: HumanPlayer picks suspect and weapon from numbered menus, room is automatic.
- Implemented clockwise disproving: iterates through other players, first match shows one card.
- Implemented accusation flow: HumanPlayer warned it is their only accusation, checks against envelope.
- Implemented correct accusation win condition: game ends, solution printed.
- Implemented wrong accusation elimination: player marked inactive, stays available for disproving.
- Verified suggestion flow works end-to-end with console output.

Blockers / issues:

- None.

Planned next steps:

- Improve AI player. Add DetectiveNotepad integration. Write more JUnit tests.

Entry 18 - 02 April 2026 (Thursday)

Date	Hours	Focus
02 April 2026 (Thursday)	2.5 hrs	AI player and console game improvements

Work completed:

- Improved AIPlayer: added DetectiveNotepad tracking of seen cards.
- AIPlayer now prefers unseen suspects and weapons when making suggestions.
- AIPlayer only makes accusations when DetectiveNotepad has deduced all three solution cards.
- Added game log output throughout: clear console messages for each turn phase.
- Implemented Main.java: console entry point asking for player counts and names.
- Ran full AI-only game to verify 200-turn cap and game-over handling.

Blockers / issues:

- None.

Planned next steps:

- Expand JUnit test suite. Sprint 2 stabilisation.

Entry 19 - 05 April 2026 (Sunday)

Date	Hours	Focus
05 April 2026 (Sunday)	3 hrs	JUnit expansion and Sprint 2 stabilisation

Work completed:

- Wrote BoardTest.java: 11 tests covering room detection, door handling, blocked squares, movement validation and secret passages.
- Wrote MovementTest.java: 8 tests covering adjacency, valid movement, blocked movement and room entry.
- Wrote SuggestionTest.java: 9 tests covering suggestion construction and disproving behaviour.
- Wrote GameFlowTest.java: 9 tests covering turn order, accusations, elimination and game-over.
- Total test suite expanded to 48 JUnit tests - all passing.
- Ran console playtests to validate end-to-end turn flow.

Blockers / issues:

- None.

Planned next steps:

- Sprint 2 review. Tag release. Begin Sprint 3 planning.

Entry 20 - 09 April 2026 (Thursday)

Date	Hours	Focus
09 April 2026 (Thursday)	2 hrs	Sprint 2 review and handoff planning

Work completed:

- Committed Sprint 2 complete: 48 tests passing.
- Tagged GitHub release v0.2-sprint2.
- Merged feature/sprint2 into dev and dev into main.
- Created feature/sprint3 branch off dev.
- Wrote post-sprint review for Sprint 2.
- Sent group chat message to team: Sprint 1 and 2 complete, starting Sprint 3, Monday cutoff to respond.
- Insik and Omar responded and agreed to contribute to Sprint 3.

Blockers / issues:

- None.

Planned next steps:

- TA meeting Monday. Begin Sprint 3 JavaFX GUI.

Entry 21 - 12 April 2026 (Sunday)

Date	Hours	Focus
12 April 2026 (Sunday)	1 hr	Other coursework and team follow-up

Work completed:

- Focused primarily on other module coursework this week.
- Updated risk register to reflect continued team engagement issues.
- Drafted Sprint 3 plan ready for TA meeting.

Blockers / issues:

- Other coursework deadlines. Team still not contributing despite commitment.

Planned next steps:

- TA meeting Monday. Begin Sprint 3 coding.

Entry 22 - 15 April 2026 (Wednesday)

Date	Hours	Focus
15 April 2026 (Wednesday)	1 hr	TA / lecturer contact and Sprint 3 planning

Work completed:

- Met with teaching assistant to discuss project progress and team situation.
- TA confirmed awareness of team issues - advised to carry on and document contributions accurately for peer review.
- Finalised Sprint 3 plan: JavaFX GUI as primary goal.
- Prepared Sprint 3 document ready for implementation.
- Downloaded JavaFX SDK 21 from Gluon and set up in IntelliJ.

Blockers / issues:

- None.

Planned next steps:

- Start Sprint 3: JavaFX setup and board display.

Entry 23 - 16 April 2026 (Thursday)

Date	Hours	Focus
16 April 2026 (Thursday)	1 hr	Sprint 3 kick-off meeting and team re-engagement

Work completed:

- Met with Insik and Omar to review the current state of the project after Sprints 1 and 2
- Gave an overview of the progress completed so far, including the working console prototype, 48 passing JUnit tests and the current GitHub repo state
- Ran a short live demo of the console version to show the gameplay and the current status
- Discussed the remaining sprint 3 work: JavaFX GUI, AI improvements, testing expansion, and final documentation.
- Agreed that Insik would focus on Ai improvements and that Omar would assist with testing and documentation

Blockers / issues:

- Team members had been out of contact for a month before this meeting, so there was limited time remaining before the deadline

Planned next steps:

- Begin sprint 3 JavaFX GUI implementation
- Monitor whether the agreed Ai and documentation tasks are actually completed

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Entry 24 - 19 April 2026 (Sunday)

Date	Hours	Focus
19 April 2026 (Sunday)	3 hrs	Sprint 3 start - JavaFX setup and board display

Work completed:

- Created feature/sprint3 branch off dev.
- Created ClueGameApp.java: JavaFX Application entry point opening a blank Clue! window.

- Created BoardView.java: loads board.png from classpath, overlays Canvas for player tokens.
- Calculated pixel-to-grid mapping from board image dimensions (1296x1226, 25 rows x 24 cols).
- Player tokens rendered as coloured circles at correct grid positions.
- Console Main.java kept working alongside GUI entry point.

Blockers / issues:

- None.

Planned next steps:

- Add sidebar, dice roll button and basic game interaction.

Entry 25 - 22 April 2026 (Wednesday)

Date	Hours	Focus
22 April 2026 (Wednesday)	4 hrs	GUI sidebar, controls and movement interaction

Work completed:

- Created GameController.java: BorderPane layout with BoardView centre and right sidebar.
- Created SidebarView.java: player name, hand of cards, Roll Dice button, game log text area.
- Wired Roll Dice button to existing Game model via GameController.
- Implemented click-to-move: clickable grid overlay, valid moves highlighted, token updates on click.
- Implemented room entry: click on door to enter, token moves to room, sidebar updates.
- Implemented AI turns: AI moves and acts automatically, GUI updates after each AI action.

Blockers / issues:

- Board pixel alignment required several iterations to match the image correctly.

Planned next steps:

- Implement suggestion and accusation dialogs. Privacy screen.

Entry 26 - 24 April 2026 (Friday)

Date	Hours	Focus
24 April 2026 (Friday)	4 hrs	Suggestion, accusation dialogs and privacy screen

Work completed:

- Implemented suggestion dialog: ChoiceBox dropdowns for suspect and weapon, room shown automatically.
- Implemented disproving display: shown privately to suggesting player only.
- Implemented accusation dialog with clear warning: this is your only accusation.
- Implemented privacy / pass screen between human turns: hides previous player hand.
- Created SetupView.java: player count selection, name entry, character assignment.

- Created NotebookView.java: manual detective notebook with checkboxes for all suspects, weapons and rooms.
- Rules alignment fixes: Miss Scarlett goes first, non-player suspect pieces placed in rooms, weapon movement on suggestions.
- Expanded test suite to 85 JUnit tests: all passing.

Blockers / issues:

- None.

Planned next steps:

- Documentation phase: sprint docs, design diagrams, group report, video.

Entry 27 - 25 April 2026 (Saturday)

Date	Hours	Focus
25 April 2026 (Saturday)	3 hrs	Final gameplay fixes, AI updates and test suite completion

Work completed:

- Completed final rules alignment fixes: character-selected starts, non-player suspect pieces placed in rooms at setup, weapon movement on suggestions verified.
- Updated AIPlayer to handle GUI turn flow reliably: fixed edge cases where AI could stall mid-turn.
- Added BoardLayoutTest.java: 10 tests covering final board dimensions, door positions, blocked edge regions, staircase blocking and start-square access.
- Added EliminationTest.java: 5 tests covering elimination rules, continued disproving after elimination and game-ending conditions.
- Added GameControllerTest.java: 11 tests covering movement state, room exits, suggestion handling, accusation handling and turn advancement.
- Full automated test suite reached 85 JUnit tests: all passing.
- Ran full manual playtests of the JavaFX GUI: setup, movement, room entry, suggestions, disproving, AI turns and game-over all verified.
- Committed final Sprint 3 code. Tagged v0.3-sprint3.

Blockers / issues:

- None.

Planned next steps:

- Documentation phase: sprint docs, testing doc, design doc, dev log, group report.

Entry 28 - 26 April 2026 (Sunday)

Date	Hours	Focus

26 April 2026 (Sunday)	5 hrs	Documentation - sprint docs, testing doc, design doc and dev log

Work completed:

- Completed Sprint 1 document: updated task statuses, added user stories, testing section with FR links and post-sprint review.
- Completed Sprint 2 document: updated task statuses, added testing section with FR links and post-sprint review.
- Completed Sprint 3 document: full record of GUI implementation, testing expansion and post-sprint review.
- Produced testing documentation: testing strategy, 85-test coverage summary, requirements traceability table and manual system testing record.
- Produced design documentation: high-level architecture, package structure, class descriptions, key design decisions, OO principles applied and behavioural design notes.
- Uploaded documentation to GitHub docs/ folder.

Blockers / issues:

- None.

Planned next steps:

- Group report, peer assessment, video recording and final submission preparation.

Entry 29 - 27 April 2026 (Monday)

Date	Hours	Focus
27 April 2026 (Monday)	4 hrs	Final document refinement and submission-pack preparation

Work completed:

- Reviewed and revised the group report, sprint documents, testing documentation and design documentation.
- Cleaned up document wording, contribution summaries and peer review sections to align with the final submission position.
- Generated JavaDocs for the codebase and added them to the documentation set.

Blockers / issues:

- Final documentation polish is still ongoing.

Planned next steps:

- Finalise peer review section, complete final report wording pass and prepare remaining submission materials.

Entry 30 - 28 April 2026 (Tuesday)

Date	Hours	Focus

28 April 2026 (Tuesday)	4 hrs	Final submission checkpoint, peer review preparation and final documentation pass

Work completed:

- Reviewed final gameplay logs to confirm that AI turn flow, accusation handling and endgame behaviour were stable enough for submission.
- Added the peer review allocation table to the group report.
- Performed a final pass on the design documentation to align diagram references with the actual files included in the submission pack.
- Created a Sprint 3 checkpoint source commit: f606c5d with message Sprint 3 checkpoint: UI, AI and submission prep updates.

Blockers / issues:

- Peer review responses and final video still outstanding. Final documentation wording still being polished.

Planned next steps:

- Complete peer review follow-up, final video, final documentation commit and submission packaging.